

3-1 · Understand Ratios

Connect & apply · 8 tasks · Prepares for 6.RP.A.1

Use the explanation to discuss strategies. Accept mathematically equivalent answers.

1. FOUNDATION 1

Compute 8×7 .

56

Use $8 \times (5 + 2) = 40 + 16$.

Skill: Recall multiplication facts through 12×12

2. FOUNDATION 2

Continue the pattern: 6, 12, 18, ____.

24

Each step adds 6.

Skill: Skip-count by any one-digit number

3. FOUNDATION 3

There are 5 blue counters and 7 red counters.
How many counters in all?

12

Add both parts: $5 + 7 = 12$.

Skill: Count each part of a part-whole picture, then add the parts for the total

4. FOUNDATION 4

Three of eight equal slices are shaded. Write the shaded fraction.

$\frac{3}{8}$

The numerator counts shaded slices; the denominator counts all equal slices.

Skill: Write a fraction naming a part of a whole

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5. CHECK A

6×4 ? There are 3 dogs and 5 cats — how many animals in all?

$6 \times 4 = 24$. Total animals: $3 \text{ dogs} + 5 \text{ cats} = 8$ animals total. (Part-to-part ratio is $3 : 5$; part-to-whole ratio of dogs is $3 : 8$).

Compare each part of your work with the worked solution.

6. CHECK B

7×3 ? There are 4 red marbles and 6 blue — how many marbles in all?

$7 \times 3 = 21$. Total: $4 + 6 = 10$ marbles. Part-to-part is $4 : 6$; red to total is $4 : 10$.

Compare each part of your work with the worked solution.

7. CHECK C

8×5 ? A team has 5 guards and 7 forwards — how many players in all?

$8 \times 5 = 40$. Total: $5 + 7 = 12$ players. Guards to forwards is $5 : 7$; guards to total is $5 : 12$.

Compare each part of your work with the worked solution.

8. TRANSFER

In a class the ratio of students wearing sneakers to boots is $5 : 2$. There are 28 students. How many wear sneakers?

$5 + 2 = 7$ parts, and $28 \div 7 = 4$ students per part, so sneakers = $5 \times 4 = 20$.

Explain why your method works and include units when needed.

3-2 · Rates and Unit Rates

Connect & apply · 8 tasks · Prepares for 6.RP.A.2

Use the explanation to discuss strategies. Accept mathematically equivalent answers.

1. FOUNDATION 1

Compute $168 \div 12$.

14

$12 \times 10 = 120$ and $12 \times 4 = 48$; together they make 168.

Skill: Divide a multi-digit number by a one- or two-digit number

2. FOUNDATION 2

Five notebooks cost \$15. What is the cost in dollars for one notebook?

3

Divide cost by the number of notebooks: $15 \div 5 = 3$ dollars per notebook.

Skill: Divide to find the amount for one, and read "per" as "for each one"

3. FOUNDATION 3

A table gives 1 bag $\rightarrow$ 4 marbles, 2 bags $\rightarrow$ 8, 3 bags $\rightarrow$ 12. How many marbles for 5 bags?

20

Each bag adds 4 marbles; $5 \times 4 = 20$.

Skill: Keep a repeating pattern going in a table

4. FOUNDATION 4

A cyclist rides 24 miles in 2 hours. State the unit rate with its unit.

12 miles per hour

Divide miles by hours: $24 \div 2 = 12$ miles per hour.

Skill: Attach the correct unit label to a computed number

3-2 · Rates and Unit Rates

Connect & apply · 8 tasks · Prepares for 6.RP.A.2

Use the explanation to discuss strategies. Accept mathematically equivalent answers.

5. CHECK A

$60 \div 4$? If 4 pencils cost \$2, what does 1 pencil cost?

$60 \div 4 = 15$. Unit price: 1 pencil costs $\$2.00 \div 4 = \0.50 (50 cents).

Compare each part of your work with the worked solution.

6. CHECK B

$72 \div 6$? If 6 notebooks cost \$9, what does 1 notebook cost?

$72 \div 6 = 12$. Unit price: $\$9 \div 6 = \1.50 per notebook.

Compare each part of your work with the worked solution.

7. CHECK C

$150 \div 5$? If a car travels 150 miles in 5 hours, what is the speed per hour?

$150 \div 5 = 30$. Speed: 30 miles per hour.

Compare each part of your work with the worked solution.

8. TRANSFER

Brand A is 12 oz for \$3.00. Brand B is 20 oz for \$4.60. Which is the better buy?

A: $\$3.00 \div 12 = \0.25 per oz. B: $\$4.60 \div 20 = \0.23 per oz. Brand B is the better buy.

Explain why your method works and include units when needed.

3-3 · Equivalent Ratios Using Tables

Connect & apply · 8 tasks · Prepares for 6.RP.A.3.a

Use the explanation to discuss strategies. Accept mathematically equivalent answers.

1. FOUNDATION 1

Scale the pair 3 : 5 by a factor of 4.

12 : 20

Multiply both terms by 4: $3 \times 4 = 12$ and $5 \times 4 = 20$.

Skill: Multiply both numbers of a pair by the same factor

2. FOUNDATION 2

Compute 9×8 .

72

$10 \times 8 - 8 = 80 - 8 = 72$.

Skill: Recall multiplication facts through 12×12

3. FOUNDATION 3

Fill in the numerator: $2/3 = \underline{\hspace{1cm}}/15$.

10

The denominator is multiplied by 5, so multiply the numerator by 5 too.

Skill: Write equivalent fractions by multiplying numerator and denominator

4. FOUNDATION 4

A table lists (2, 6), (4, 12), (6, 18). What pair comes next if the same step continues?

(8, 24)

Add 2 to the first coordinate and 6 to the second.

Skill: Fill the next row of a growing pattern table

3-3 · Equivalent Ratios Using Tables

Connect & apply · 8 tasks · Prepares for 6.RP.A.3.a

Use the explanation to discuss strategies. Accept mathematically equivalent answers.

5. CHECK A

2×5 and 3×5 ? Complete: $2 : 3 = 8 : \underline{\hspace{1cm}}$

$2 \times 5 = 10$ and $3 \times 5 = 15$. Complete ratio: Since $2 \times 4 = 8$, scale 3 by 4: $3 \times 4 = 12$ ($2 : 3 = 8 : 12$).

Compare each part of your work with the worked solution.

6. CHECK B

4×6 and 5×6 ? Complete: $4 : 5 = 24 : \underline{\hspace{1cm}}$

$4 \times 6 = 24$ and $5 \times 6 = 30$. Since $4 \times 6 = 24$, scale 5 by 6: $4 : 5 = 24 : 30$.

Compare each part of your work with the worked solution.

7. CHECK C

3×7 and 8×7 ? Complete: $3 : 8 = 21 : \underline{\hspace{1cm}}$

$3 \times 7 = 21$ and $8 \times 7 = 56$. So $3 : 8 = 21 : 56$.

Compare each part of your work with the worked solution.

8. TRANSFER

A recipe uses 3 cups flour to 2 cups water. You have 8 cups of water. How much flour?

Water scaled by 4 ($2 \times 4 = 8$), so flour is $3 \times 4 = 12$ cups.

Explain why your method works and include units when needed.

3-4 · Equivalent Ratios Using Graphs

Connect & apply · 8 tasks · Prepares for 6.RP.A.3.a

Use the explanation to discuss strategies. Accept mathematically equivalent answers.

1. FOUNDATION 1

Describe how to plot (3, 5) from the origin.

Move 3 units right and 5 units up

The first coordinate is horizontal; the second is vertical.

Skill: Plot a point in Quadrant I from an ordered pair

2. FOUNDATION 2

The x-axis ticks are 0, 2, 4, 6, 8. What x-value is at the third tick to the right of 0?

6

Three intervals of 2 give 6.

Skill: Read the scale on each axis before plotting

3. FOUNDATION 3

A point is 4 units right and 7 units up from the origin. Give its ordered pair.

(4, 7)

Write horizontal position first, then vertical position.

Skill: Count right from the origin, then up

4. FOUNDATION 4

The rule is $y = 3x$. Give y when $x = 4$.

12

Substitute 4 for x : $3 \times 4 = 12$.

Skill: Build a ratio table to generate the pairs

3-4 · Equivalent Ratios Using Graphs

Connect & apply · 8 tasks · Prepares for 6.RP.A.3.a

Use the explanation to discuss strategies. Accept mathematically equivalent answers.

5. CHECK A

Start at (0, 0), move 3 right and 2 up — write the ordered pair. If each gridline is 5, where is 15?

Ordered pair: (3, 2) — move 3 units right on x-axis, then 2 units up on y-axis. Grid position: If each grid line is 5, 15 is at line 3 ($15 \div 5 = 3$).

Compare each part of your work with the worked solution.

6. CHECK B

Start at (0, 0), move 5 right and 4 up — write the ordered pair. If each gridline is 2, where is 12?

Ordered pair (5, 4). With gridlines worth 2, the value 12 sits at line 6 ($12 \div 2 = 6$).

Compare each part of your work with the worked solution.

7. CHECK C

Start at (0, 0), move 2 right and 7 up — write the ordered pair. If each gridline is 10, where is 50?

Ordered pair (2, 7). With gridlines worth 10, the value 50 sits at line 5 ($50 \div 10 = 5$).

Compare each part of your work with the worked solution.

8. TRANSFER

A ratio graph passes through (0, 0), (2, 6), and (4, 12). What is the unit rate, and where is $x = 5$?

Unit rate = $6 \div 2 = 3$, so $y = 3x$. At $x = 5$, $y = 15$ — the point (5, 15).

Explain why your method works and include units when needed.

3-5 · Comparing Ratio Relationships

Connect & apply · 8 tasks · Prepares for 6.RP.A.3.b

Use the explanation to discuss strategies. Accept mathematically equivalent answers.

1. FOUNDATION 1

Which is greater: 35 or 53?

53

Compare tens: 5 tens is greater than 3 tens.

Skill: Compare two whole numbers

2. FOUNDATION 2

Scale 4 : 7 to an equivalent ratio with first term 20.

20 : 35

Multiply both terms by 5.

Skill: Scale both parts of a ratio to hit a target value

3. FOUNDATION 3

What is the least common multiple of 6 and 8?

24

Multiples of 6 are 6, 12, 18, 24; multiples of 8 are 8, 16, 24.

Skill: Find a common multiple of two numbers

4. FOUNDATION 4

Pack A costs \$12 for 4 pens. Pack B costs \$14 for 7 pens. Which has the lower price per pen?

Pack B

A costs \$3 per pen; B costs \$2 per pen.

Skill: Divide to find a unit rate for a fair comparison

3-5 · Comparing Ratio Relationships

Connect & apply · 8 tasks · Prepares for 6.RP.A.3.b

Use the explanation to discuss strategies. Accept mathematically equivalent answers.

5. CHECK A

Which is greater, 12 or 9? Scale 3 : 4 so the first number is 12.

12 is greater than 9. Scale 3 : 4: Since $3 \times 4 = 12$, multiply 4 by 4 to get 16. Scaled ratio: 12 : 16.

Compare each part of your work with the worked solution.

6. CHECK B

Which is greater, 18 or 15? Scale 2 : 5 so the first number is 18.

18 is greater. Since $2 \times 9 = 18$, multiply 5 by 9: $2 : 5 = 18 : 45$.

Compare each part of your work with the worked solution.

7. CHECK C

Which is greater, 24 or 30? Scale 4 : 7 so the first number is 24.

30 is greater. Since $4 \times 6 = 24$, multiply 7 by 6: $4 : 7 = 24 : 42$.

Compare each part of your work with the worked solution.

8. TRANSFER

Runner A covers 3 miles in 24 minutes; Runner B covers 5 miles in 45 minutes. Who is faster?

A: $24 \div 3 = 8$ minutes per mile. B: $45 \div 5 = 9$ minutes per mile. Runner A is faster (fewer minutes per mile).

Explain why your method works and include units when needed.

3-6 · Converting Within One System

Connect & apply · 8 tasks · Prepares for 6.RP.A.3.d

Use the explanation to discuss strategies. Accept mathematically equivalent answers.

1. FOUNDATION 1

Compute $72 \div 9$.

8

Check with $9 \times 8 = 72$.

Skill: Recall multiplication and division facts

2. FOUNDATION 2

Use 1 foot = 12 inches. How many inches are in 3 feet?

36

Multiply $3 \times 12 = 36$ inches.

Skill: Recall customary facts: 1 ft = 12 in, 1 lb = 16 oz, 1 gal = 4 qt

3. FOUNDATION 3

How many meters are in 2.4 kilometers? Use 1 km = 1000 m.

2400

Multiply 2.4 by 1000 because each kilometer contains 1000 meters.

Skill: Multiply and divide by 10, 100, 1000 for metric steps

4. FOUNDATION 4

Convert 48 inches to feet using 12 inches = 1 foot.

4

Divide by 12 because 12 inches form one foot.

Skill: Decide whether a conversion calls for multiplying or dividing

3-6 · Converting Within One System

Connect & apply · 8 tasks · Prepares for 6.RP.A.3.d

Use the explanation to discuss strategies. Accept mathematically equivalent answers.

5. CHECK A

12×4 ? 1 foot = ____ inches? 2.5 m = ____ cm?

$12 \times 4 = 48$. 1 foot = 12 inches. 2.5 m = 250 cm ($2.5 \times 100 = 250$).

Compare each part of your work with the worked solution.

6. CHECK B

11×4 ? 1 pound = ____ ounces? 3.5 m = ____ cm?

$11 \times 4 = 44$. 1 pound = 16 ounces. 3.5 m = 350 cm (3.5×100).

Compare each part of your work with the worked solution.

7. CHECK C

9×6 ? 1 gallon = ____ quarts? 0.75 km = ____ m?

$9 \times 6 = 54$. 1 gallon = 4 quarts. 0.75 km = 750 m (0.75×1000).

Compare each part of your work with the worked solution.

8. TRANSFER

A tank holds 3 gallons. How many cups is that? (1 gal = 4 qt, 1 qt = 4 cups)

$3 \text{ gal} \times 4 = 12 \text{ qt}$, and $12 \text{ qt} \times 4 = 48 \text{ cups}$.

Explain why your method works and include units when needed.

3-7 · Converting Between Systems

Connect & apply · 8 tasks · Prepares for 6.RP.A.3.d

Use the explanation to discuss strategies. Accept mathematically equivalent answers.

1. FOUNDATION 1

Compute 6×2.5 .

15

$$6 \times 2 + 6 \times 0.5 = 12 + 3.$$

Skill: Multiply a whole number by a decimal

2. FOUNDATION 2

Round 7.846 to the nearest hundredth.

7.85

The thousandths digit is 6, so round the hundredths digit up.

Skill: Round a decimal to the nearest tenth or hundredth

3. FOUNDATION 3

Use 1 inch = 2.54 centimeters. Convert 5 inches to centimeters.

12.7

$$5 \times 2.54 = 12.70 \text{ centimeters.}$$

Skill: Multiply by a conversion factor and keep the unit label

4. FOUNDATION 4

Use 1 yard = 3 feet and 1 foot = 12 inches. Convert 2 yards to inches.

72

First $2 \times 3 = 6$ feet, then $6 \times 12 = 72$ inches.

Skill: Chain two conversions in order without losing a step

3-7 · Converting Between Systems

Connect & apply · 8 tasks · Prepares for 6.RP.A.3.d

Use the explanation to discuss strategies. Accept mathematically equivalent answers.

5. CHECK A

4×1.6 ? 10×0.45 ? Round 6.44 to the nearest tenth.

$4 \times 1.6 = 6.4$. $10 \times 0.45 = 4.5$. Round 6.44 to nearest tenth = 6.4 (since the hundredths digit 4 is less than 5).

Compare each part of your work with the worked solution.

6. CHECK B

5×2.54 ? 8×0.45 ? Round 12.7 to the nearest tenth.

$5 \times 2.54 = 12.7$. $8 \times 0.45 = 3.6$. 12.7 is already to the nearest tenth: 12.7.

Compare each part of your work with the worked solution.

7. CHECK C

3×1.6 ? 20×0.45 ? Round 9.08 to the nearest tenth.

$3 \times 1.6 = 4.8$. $20 \times 0.45 = 9.0$. Rounding 9.08 to the nearest tenth gives 9.1.

Compare each part of your work with the worked solution.

8. TRANSFER

A race is 5 kilometers. About how many miles is that, if $1 \text{ km} \approx 0.62 \text{ mi}$? Round to the nearest tenth.

$5 \times 0.62 = 3.1 \text{ miles}$.

Explain why your method works and include units when needed.